BUSINESS CASE

AI-Powered Country Risk Assessment

Automating Structured Risk Scoring from IMF Macroeconomic Data

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Executive Summary

This document presents a business case for deploying an LLM-powered country risk assessment pipeline within a financial services organization. The pipeline ingests IMF World Economic Outlook data and generates structured, auditable risk scores across four categories - economic, financial, political, and operational. Demonstrated across 12 representative economies, the architecture is designed to scale to full IMF WEO coverage of 190+ countries.

A working prototype has been built and validated. This document outlines the problem, the solution design, governance considerations, and what a path to production would involve

Section 1 - The Business Problem

Why Country Risk Assessment Needs to Change

Country risk assessment is a core function in asset management and investment banking. Portfolio managers, sovereign debt analysts, and ESG teams need to understand macroeconomic stability across dozens of countries simultaneously, and that assessment needs to be consistent, repeatable, and fast.

Today, the process is largely manual. Analysts open IMF data tables, review indicators individually, apply their own judgment, and write narratives for stakeholders. This works but it does not scale, it is inconsistent across teams, and it consumes analyst time that could be spent on higher-value interpretation.

Current State (Manual)	Future State (AI-Assisted)
Analyst manually reviews IMF tables per country	Pipeline demonstrated across 12 economies architecture designed to scale to full IMF WEO coverage of 190+ countries
Inconsistent scoring across different analysts	Standardized risk framework applied uniformly every time
Hours to produce a full risk report	Executive summary generated in seconds per country
No audit trail on how scores were derived	Full rationale and key signals logged in every output
Data quality issues discovered too late	Built-in data quality flagging before scores are used
Outputs siloed in analyst spreadsheets	Clean CSV feeds directly into Power BI or Tableau

Section 2 - The Solution

AI-Powered Risk Scoring Pipeline

This project demonstrates a working LLM pipeline that automates country risk scoring from raw IMF macroeconomic data. Built using Python, the Groq API with LLaMA 3.3 70B, and pandas - it outputs structured CSV data ready for downstream BI tools.

How the Pipeline Works:

- IMF World Economic Outlook data (190+ countries available, 8 macroeconomic indicators) is loaded and cleaned, pipeline demonstrated across 12 representative economies

- Key indicators selected per country: GDP growth, inflation, unemployment, fiscal balance, current account, trade volumes
- A structured prompt is sent to LLaMA 3.3 via the Groq API, framed as a senior risk analyst assessment
- The model returns a structured JSON object with scores, signals, executive summary, and a data quality flag
- Outputs are flattened to CSV and fed into Power BI or Tableau for visualisation

What the Pipeline Produces Per Country:

Output Field	Description
Overall Risk Level	Low / Medium / High / Critical - top-line rating for each country
Risk Score (x4)	1–10 score across Economic, Financial, Political, Operational categories
Key Signals	3 specific data points driving each category score
Executive Summary	3-sentence stakeholder narrative ready to paste into a report
Recommended Action	Monitor / Caution / Restrict / Avoid clear, actionable recommendation
Data Quality Flag	Highlights missing, extreme, or unreliable indicators before scores are used

Section 3 - Responsible AI & Governance

Governance Framework

Deploying AI in financial services requires explicit governance. This pipeline has been designed with responsible AI principles embedded from the outset - not added as an afterthought.

Governance Area	How This Pipeline Addresses It
Transparency	All prompt logic is version-controlled and auditable. Scoring rationale is included in every output.
Data Quality	Built-in data_quality_flag field surfaces unreliable or missing indicators before scores are acted upon.
Human Oversight	Outputs are framed as decision-support inputs, not automated decisions. Analyst review is required before any action is taken.
Source Integrity	Data sourced exclusively from IMF World Economic Outlook a credible, publicly documented, internationally recognised source.
Known Limitations	Political risk is inferred from economic signals only. Geopolitical events and qualitative factors are outside the current scope and must be overlaid by the analyst.
Bias Awareness	LLM outputs can reflect training data biases. Scores should be reviewed critically, especially for countries with limited IMF data coverage.

Section 4 - Path to Production

How to Scale This in a Financial Services Organisation

Moving from prototype to production requires involvement from technology, data governance, legal, compliance, and the business teams who will use the outputs. The following phased approach is recommended:

#	Phase	What It Involves
1	Stakeholder Alignment	Identify champion teams (e.g. sovereign debt, ESG, risk). Run a working session to validate the risk framework and scoring categories against existing analyst methodology.
2	Data & Compliance Review	Confirm IMF data license for commercial use. Involve data governance and legal teams early. Document data lineage and retention policy.
3	Pilot Deployment	Run the pipeline for a focused country set relevant to the portfolio. Collect analyst feedback on score accuracy and output format. Iterate on prompt logic.
4	BI Integration	Connect CSV output to existing Power BI or Tableau infrastructure. Design dashboards with risk teams rather than for them.
5	Governance & Sign-off	Document responsible for AI framework. Define escalation path for high-risk outputs. Obtain sign-off from risk, compliance, and technology.
6	Scaled Rollout	Expand full coverage. Schedule regular pipeline runs aligned to IMF WEO update cycle (April and October). Assign ownership for ongoing maintenance.

Section 5 - What I'd Do Differently at Scale

Moving from Prototype to Production

Building this pipeline as a solo prototype was a deliberate scoping decision - but it surfaced with several important considerations that would need to be addressed before deploying this in a real financial services organization.

On scope:

- The current prototype was demonstrated across 12 representative economies spanning high, medium, and low risk profiles. Scaling to the full IMF WEO dataset of 190+ countries is technically straightforward - the pipeline architecture supports it but was deliberately scoped down for this prototype due to API rate limits and token costs on the free tier. In a production environment with a paid API plan or a self-hosted model, full coverage would be achievable in a single pipeline run.

On the data side:

- The current pipeline runs on a static IMF WEO snapshot downloaded manually. At scale, this would need to be connected to a scheduled data ingestion process aligned to IMF's April and October release cycle - removing the manual step entirely
- Data lineage and licensing would need to be formally documented. IMF data is publicly available but commercial use in a regulated firm requires sign-off from data governance and legal teams

On the model side:

- LLaMA 3.3 70B via Groq works well for a prototype but introduces dependency on a third-party API. A production deployment would need to evaluate whether to use a managed API, a self-hosted model, or a commercial provider with enterprise SLAs and data privacy guarantees
- Political risk scoring is currently inferred from economic signals only - a known limitation. At scale I would introduce a second data source (e.g. a political risk index like ICRG or PRS) to make that category more robust
- Prompt logic would need versioning and regression testing - if the prompt changes, scores change, and that needs to be tracked and auditable

On the governance side:

- The data_quality_flag field is a first step toward responsible AI but it would need to be expanded into a formal review workflow - not just a text field in a CSV
- Outputs should never feed directly into investment decisions without an analyst review step built into the process, not just stated as a guideline
- A pilot with a small analyst team would be the right next step - run the pipeline alongside existing manual processes for one quarter, compare outputs, collect feedback, and iterate before any wider rollout

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